DIGITAL IMAGE PROCESSING · LECTURE 3

Sharpening & Edge Detection

Keeping exactly what smoothing threw away

Digital Image Fundamentals · Signal Processing

ROADMAP

Today's Agenda

PART I

Derivatives in Images

First and second differences · what each responds to

PART II

The Laplacian

Second-derivative sharpening · masks · the composite operator

PART III

Unsharp Masking & Highboost

Sharpening by subtracting a blurred copy · the gain k

PART IV

The Gradient

First-derivative operators · Roberts, Prewitt, Sobel

PART V

Edge Detection

Noise and derivatives · LoG and zero crossings · Canny

PART I

Derivatives in Images

If averaging blurs, then differencing must sharpen

Smoothing and Sharpening Are Opposites

Lecture 2: smoothing

Local averaging

Keeps slow variation, removes fast variation. Blurs edges along the way.

Today: sharpening

Local differencing

Responds only where the image is changing. Flat regions give nothing.

Averaging is a kind of summation; differencing is its inverse. So the tool for sharpening is the derivative.

DEFINITIONS

Derivatives on a Pixel Grid

Pixels are discrete, so derivatives become differences between neighbouring values.

First derivative

$$\partial f/\partial x = f(x+1) - f(x)$$

Second derivative

$$\partial^2 f/\partial x^2 = f(x+1) + f(x-1) - 2f(x)$$

Zero in flat areas

Both derivatives must give zero wherever the intensity is constant.

Non-zero at a change

Both must respond at the onset of a step, a ramp, or a line.

Along a ramp

The first derivative stays non-zero all the way along a ramp; the second does not.

What Each Derivative Responds To



Read the three panels together: both derivatives are zero on flat stretches; only the first stays non-zero through the ramp; and the second produces a double response — one positive, one negative — at every step and line.

BEHAVIOUR

First vs. Second Derivative

	First derivative	Second derivative
Constant region	zero	zero
Along a ramp	non-zero throughout	zero (non-zero only at its ends)
At a step	one thick response	double response, with a zero crossing between
A thin line	responds at both sides	responds much more strongly
Typical edge produced	thicker	finer, with more detail
Sensitivity to noise	lower	higher

The second derivative is the sharper instrument — which is why sharpening starts there.

PART II

The Laplacian

The second derivative, in two dimensions

DEFINITION

The Laplacian

Add the second derivative along x to the second derivative along y.

$$\nabla^2 f = \partial^2 f/\partial x^2 + \partial^2 f/\partial y^2$$

$$\partial^2 f/\partial x^2 = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\partial^2 f/\partial y^2 = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x, y)$$

Four neighbours, minus four times the centre — that is the whole operator.

Laplacian Masks

4-neighbour Laplacian

0	1	0
1	-4	1
0	1	0

8-neighbour (includes diagonals)

1	1	1
1	-8	1
1	1	1

negated 4-neighbour

0	-1	0
-1	4	-1
0	-1	0

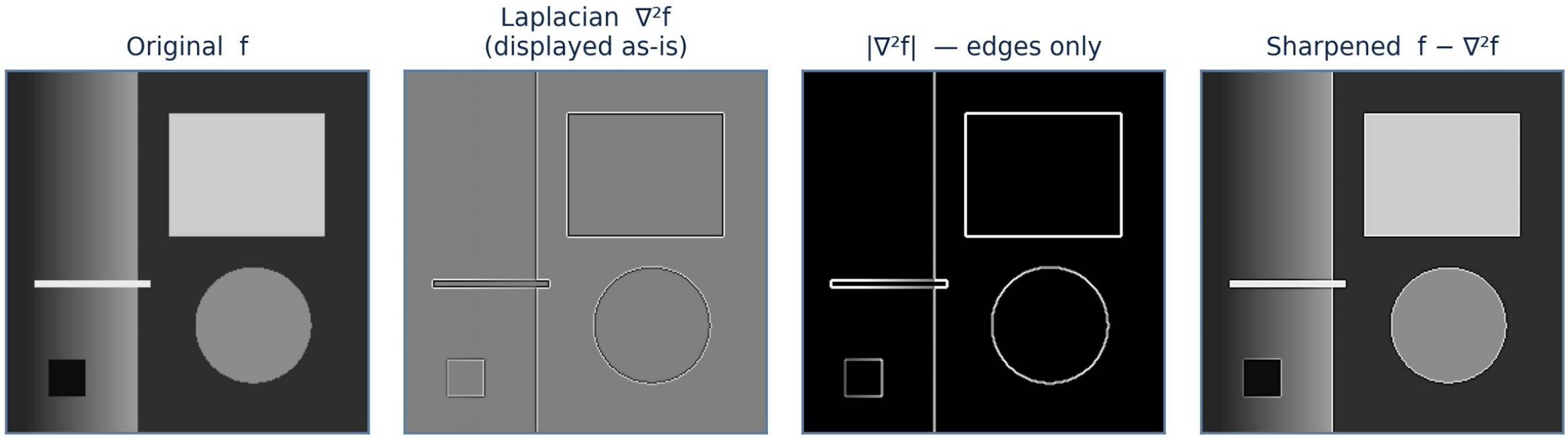
negated 8-neighbour

-1	-1	-1
-1	8	-1
-1	-1	-1

The 4-neighbour mask follows directly from the definition. Including the diagonals gives the 8-neighbour version. Both come in a negated form — the sign convention matters when we add the result back to the image.

APPLIED

The Laplacian of an Image



The Laplacian is zero across flat regions and across the smooth ramp on the left — it fires only at boundaries. Adding it back to the original restores the intensities and puts a crisp accent on every edge.

Adding the Edges Back

$$g(x, y) = f(x, y) + c \cdot \nabla^2 f(x, y)$$

c = −1

Use when the mask has a negative centre (the −4 or −8 form). Subtracting a negative-centre Laplacian brightens the edge.

c = +1

Use when the mask has a positive centre (the +4 or +8 form). Getting the sign wrong inverts the edges instead of enhancing them.

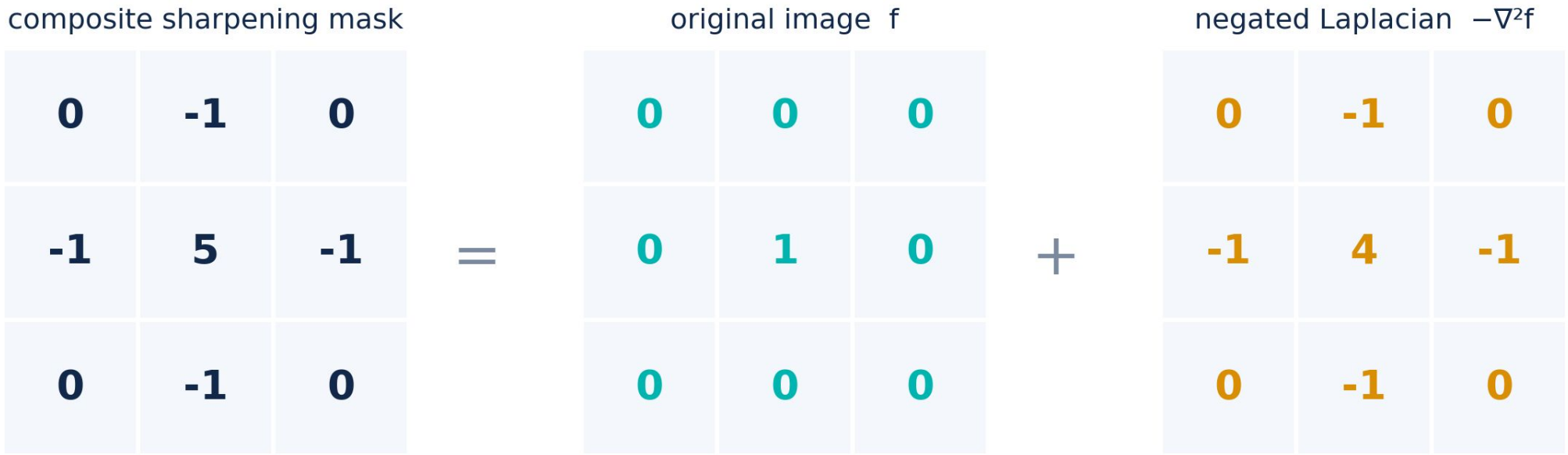
The image is preserved

Because the mask coefficients sum to zero, flat regions pass through unchanged — only the boundaries are altered.

ONE MASK INSTEAD OF TWO

The Composite Sharpening Operator

Rather than filtering and then adding, fold both steps into a single mask.



$$g(x, y) = 5f(x, y) - [\, f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) \,]$$

The 8-neighbour version gives the corresponding mask with 9 at the centre and −1 all around.

PART III

Unsharp Masking & Highboost

Sharpening without ever computing a derivative

THE IDEA

Sharpening by Subtracting a Blur

Confusingly named — "unsharp masking" is a sharpening technique. It comes from darkroom photography, where a blurred negative was used as a mask.

1. Blur

Smooth the image with any of the filters from Lecture 2 — usually a Gaussian.

2. Subtract

mask = original − blurred. What survives is precisely the detail the blur removed.

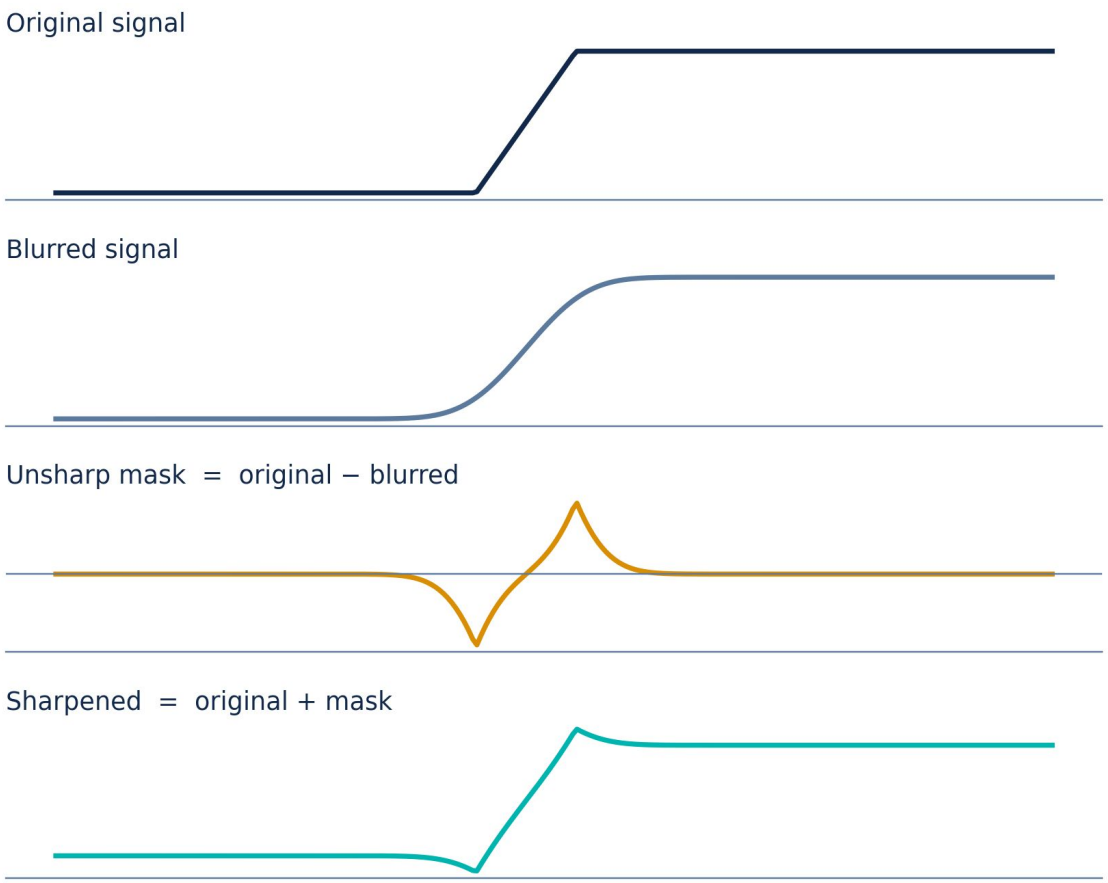
3. Add back

Add the mask to the original. The detail is now counted twice, so edges stand out.

$$g(x, y) = f(x, y) + k \cdot [\, f(x, y) - \text{blur}(x, y) \,]$$

STEP BY STEP

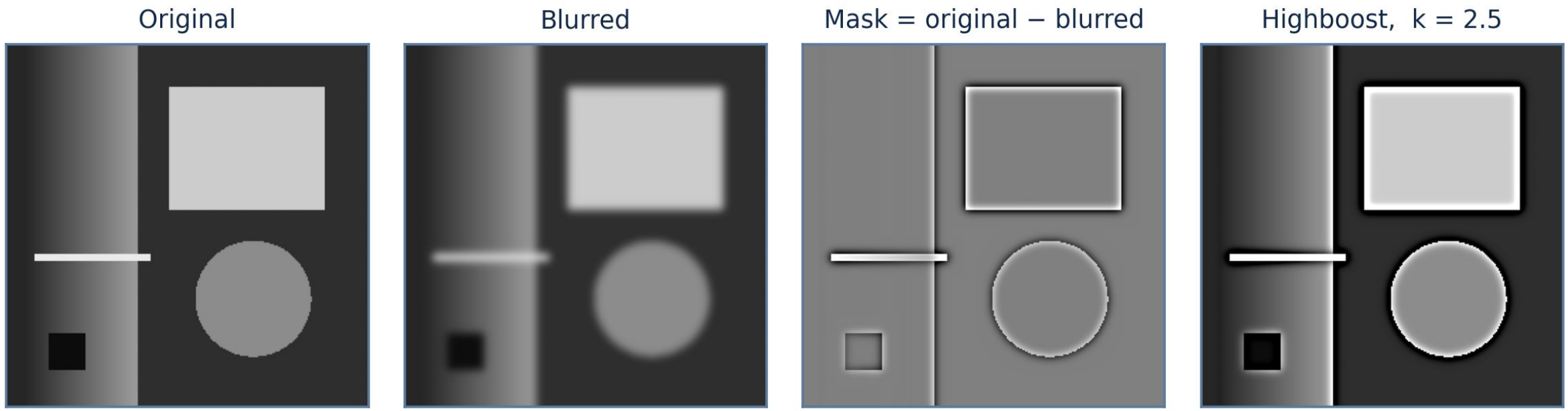
Unsharp Masking on a Single Edge



Blurring softens the edge. The difference between the two is a signal that is zero everywhere except right at the edge, where it swings negative then positive. Adding it back produces the overshoot on either side that the eye reads as sharpness.

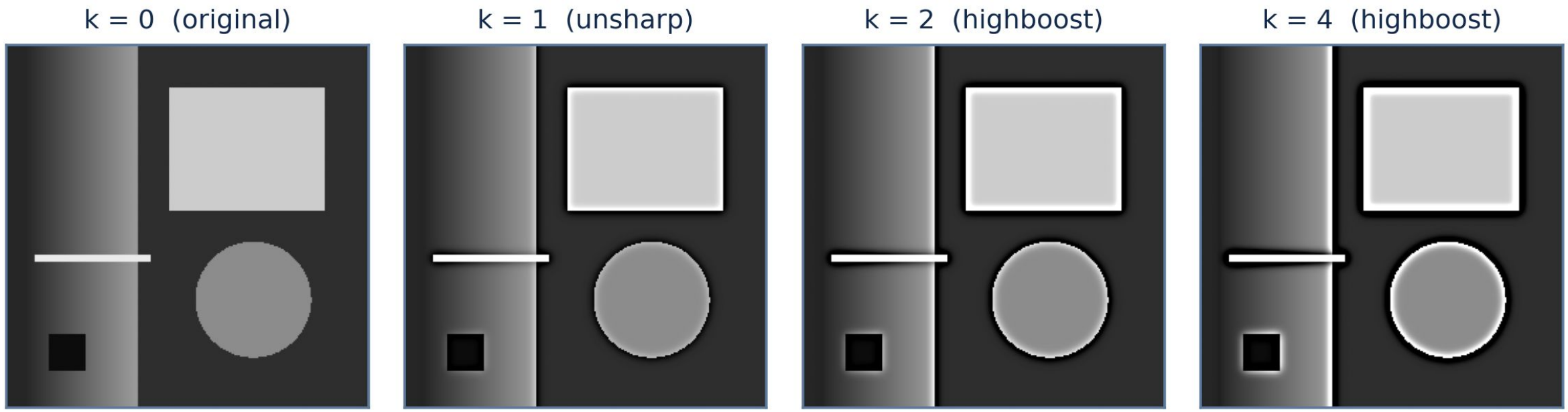
APPLIED

Unsharp Masking on an Image



The mask holds only what the blur destroyed — edges and fine structure, with mid-grey standing for zero. Scaling it up before adding it back gives highboost filtering.

Choosing k



k = 1 is ordinary unsharp masking; k > 1 is highboost filtering. Larger k means stronger edges — and, eventually, visible haloes and amplified noise. There is no single correct value; it is a judgement call.

PART IV

The Gradient

First-derivative operators and how edges are actually found

DEFINITION

The Gradient

At each pixel the first derivative is not one number but two — the rate of change along x and along y.

$$\nabla f = [\, G_x ,\ G_y \,] \quad \text{where } G_x = \partial f/\partial x,\ G_y = \partial f/\partial y$$

$$\text{magnitude} = \sqrt{(\, G_x^2 + G_y^2 \,)} \approx |G_x| + |G_y|$$

Direction matters

The gradient points across the edge — in the direction of steepest change.

Magnitude is what we display

It measures edge strength regardless of orientation, so it is the usual edge map.

Why the approximation

$|G_x| + |G_y|$ avoids squares and a square root, and is much cheaper to compute.

Roberts, Prewitt and Sobel

Roberts (2×2)

-1	0
0	1

horizontal

0	-1
1	0

vertical

smallest possible;
no centre pixel, so the
result sits between pixels

Prewitt (3×3)

-1	0	1
-1	0	1
-1	0	1

horizontal

-1	-1	-1
0	0	0
1	1	1

vertical

3×3 and centred;
all three rows weighted
equally

Sobel (3×3)

-1	0	1
-2	0	2
-1	0	1

horizontal

-1	-2	-1
0	0	0
1	2	1

vertical

3×3 and centred;
the centre line is weighted double,
which smooths as it differentiates

Each operator estimates Gx and Gy with a pair of small masks. They differ in size and in how much smoothing they build in.

A PRACTICAL POINT

Why Masks Are Usually Odd-Sized

An even mask has no centre

A 2×2 window has no middle pixel, so the value it computes belongs halfway between pixels rather than at one of them.

The result is displaced

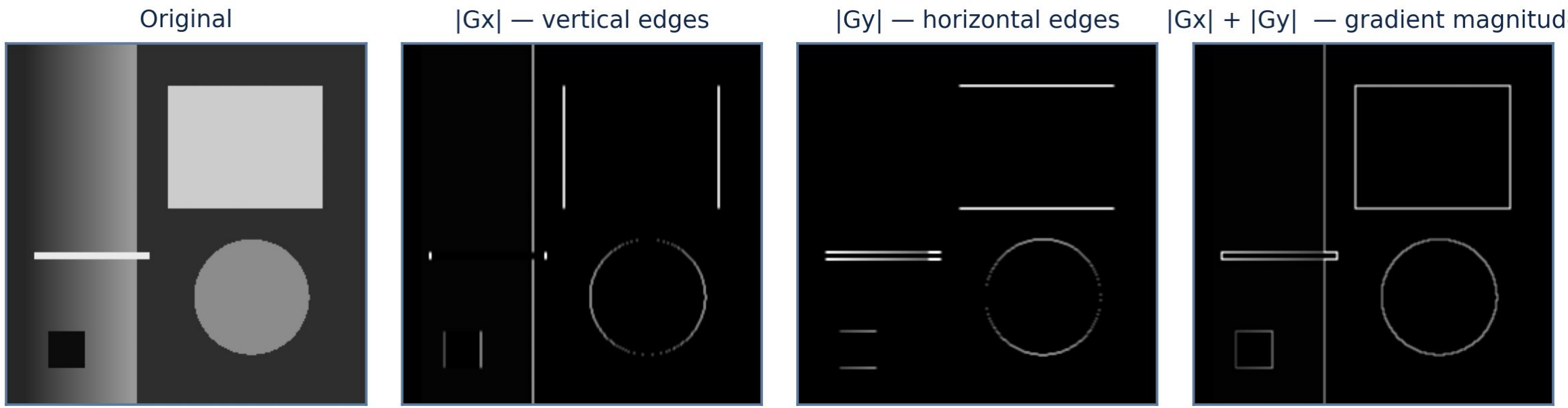
Roberts therefore produces an edge map shifted by half a pixel — tolerable, but a nuisance when locations matter.

Odd sizes are symmetric

3×3, 5×5, 7×7 all have a well-defined centre and are symmetric about it, which is why nearly every mask you meet is odd.

APPLIED

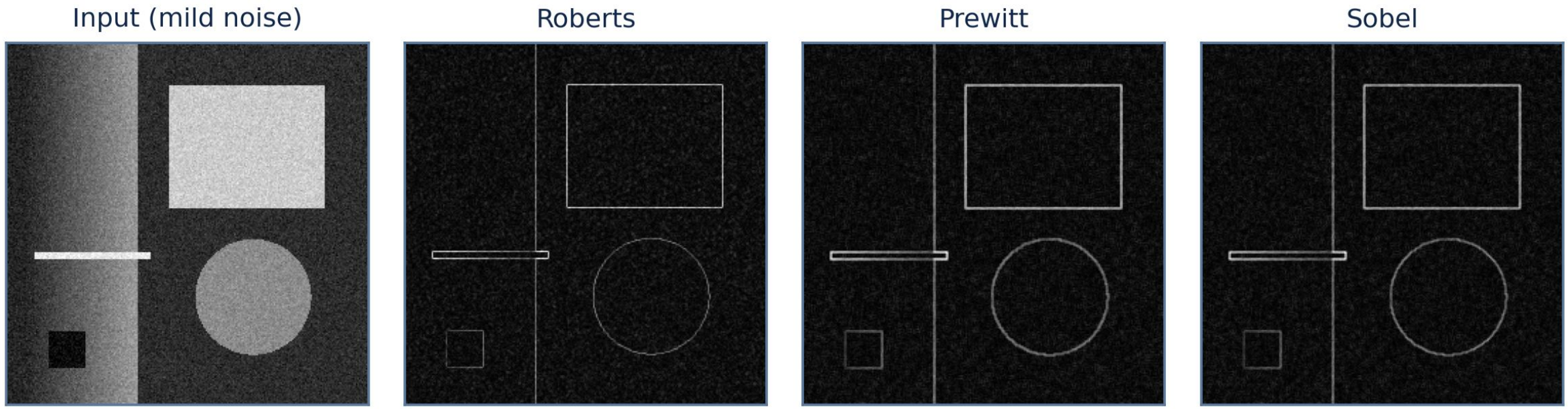
Gx, Gy and the Gradient Magnitude



The horizontal mask responds to vertical edges and the vertical mask to horizontal ones. Combining them gives an edge map that no longer depends on orientation.

COMPARISON

The Three Operators Side by Side



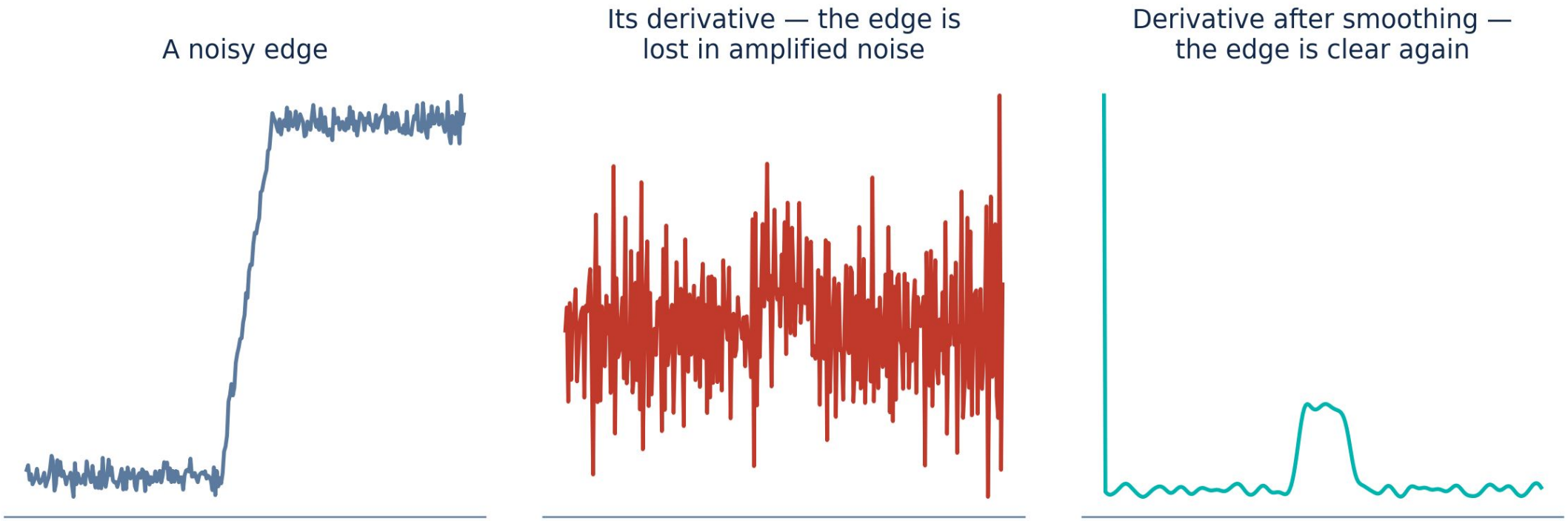
With a little noise present the differences show: Roberts is the noisiest and its edges are thinnest, Prewitt is steadier, and Sobel is cleanest — the doubled centre weight smooths while it differentiates.

PART V

Edge Detection

From an edge response to an actual edge map

Derivatives Amplify Noise



Differencing takes neighbouring values apart, so small random fluctuations become large swings. A modest amount of noise can bury the very edge we are trying to find — unless we smooth first.

THE REMEDY

Smooth First, Then Differentiate

Every serious edge detector begins with the smoothing filters from Lecture 2. The two halves of this course meet here.

Smooth

A Gaussian removes the fine fluctuations that would otherwise dominate the derivative.

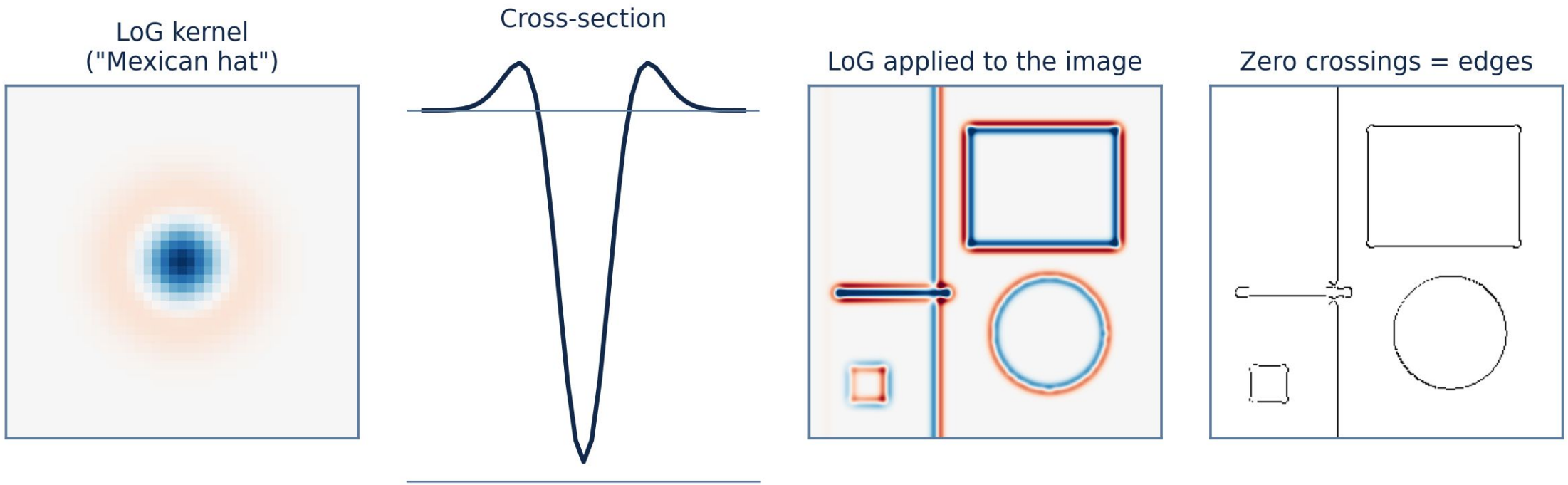
Differentiate

Apply the Laplacian or the gradient to what remains.

Decide

Convert the continuous response into a yes/no edge map by thresholding or by locating zero crossings.

Laplacian of Gaussian and Zero Crossings



Smoothing and the Laplacian can be merged into a single "Mexican hat" mask. Because the second derivative changes sign at an edge, the edge is found not where the response is large but where it crosses zero — which gives thin, closed contours.

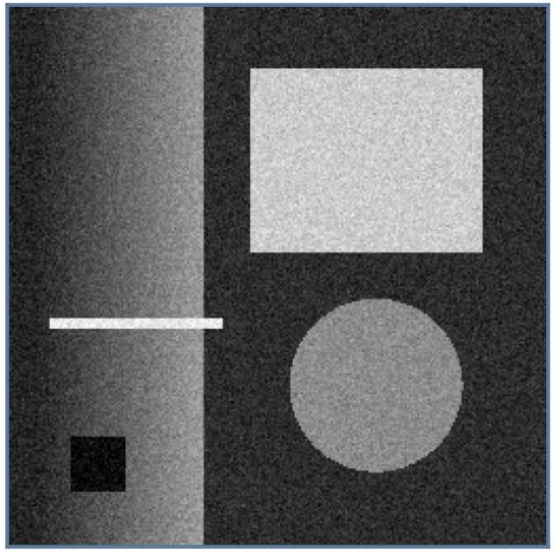
The Canny Edge Detector

The standard answer to the question "give me the edges". Five stages, each fixing a specific defect of the plain gradient.

1	**Smooth**	Gaussian filter, to control noise
2	**Gradient**	magnitude and direction at every pixel
3	**Non-maximum suppression**	keep a pixel only if it is the largest along the gradient direction — this thins wide responses to one pixel
4	**Double threshold**	strong edges accepted, weak ones held, the rest discarded
5	**Hysteresis**	keep a weak edge only if it connects to a strong one

Canny, Stage by Stage

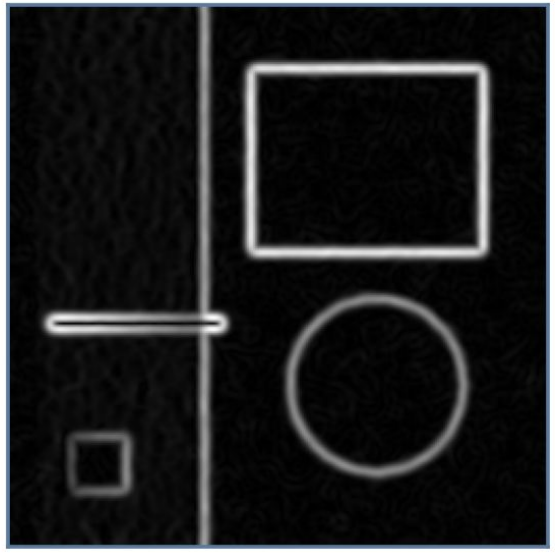
1. Input



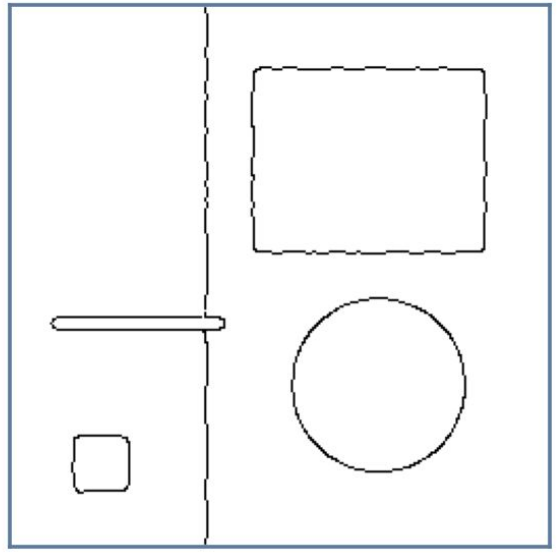
2. Smooth with a Gaussian



3. Gradient magnitude (thick responses)

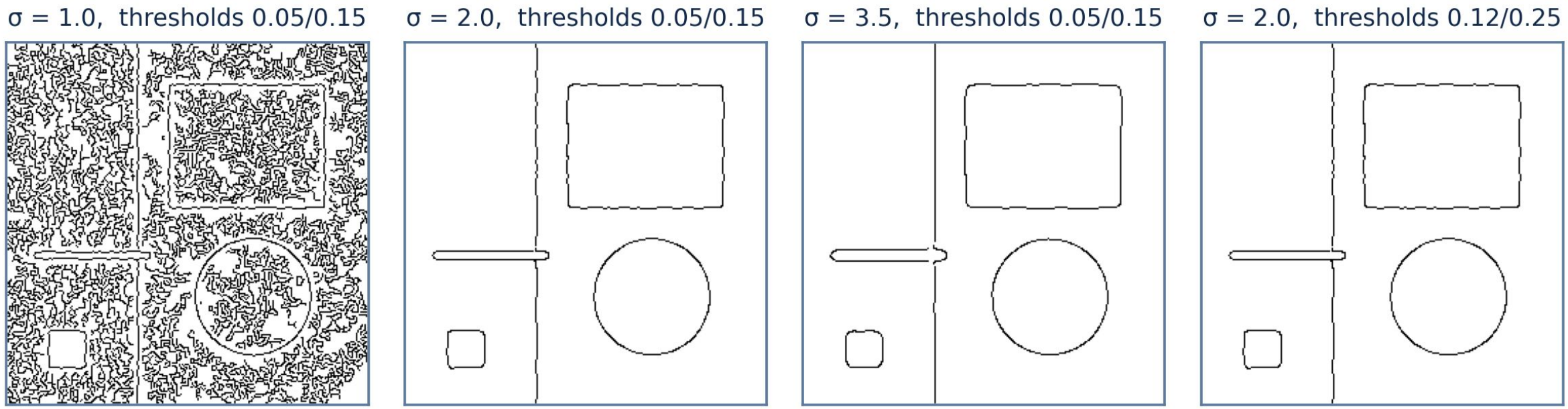


4. Thinned + hysteresis → single-pixel edges



Smoothing first, then the gradient, then thinning and hysteresis. The intermediate gradient map is thick and speckled; what comes out at the end is a set of clean single-pixel contours.

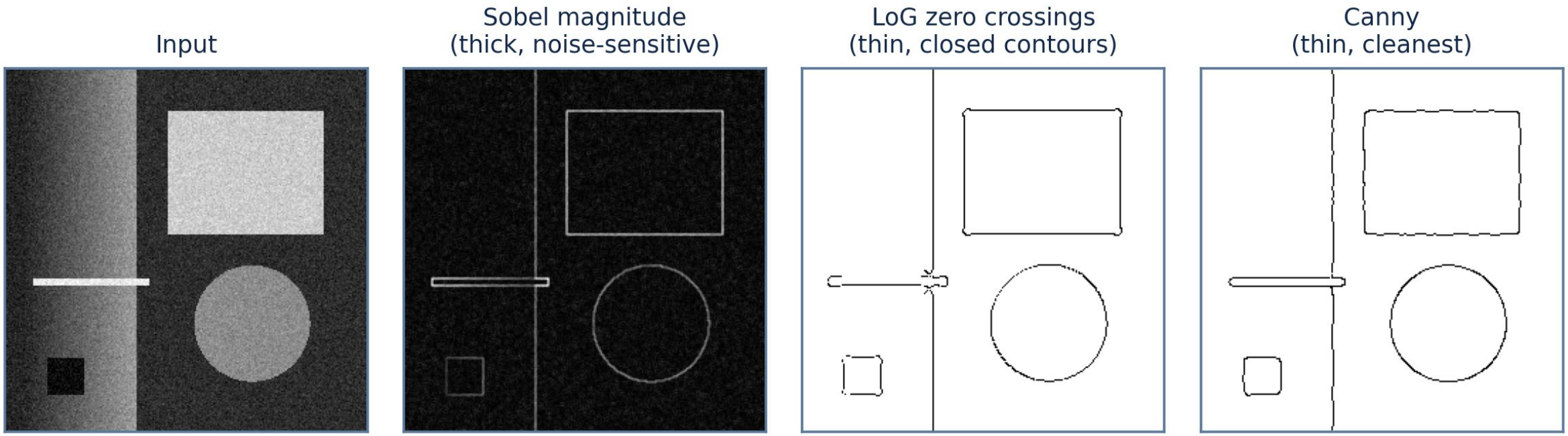
The Parameters Do the Work



A small σ keeps fine detail but admits noise; a large σ suppresses noise and rounds corners. Raising the thresholds drops the weaker contours. These choices, not the algorithm, decide what counts as an edge.

COMPARISON

Sobel vs. LoG vs. Canny



The gradient magnitude is a response, not a decision — it is thick and still carries noise. LoG gives thin closed contours but is sensitive to the threshold on the zero crossings. Canny costs the most and gives the cleanest result.

PART VI

Beyond the Textbook

Edge detectors you will meet in practice but not in Gonzalez

MOTIVATION

What Sobel and Canny Leave Open

Everything so far works. But each of these questions has a well-established answer that the textbook does not cover.

How accurate is the edge direction?	Sobel's coefficients were chosen by hand. Better ones exist.	**Scharr**
Can we get direction directly?	Instead of combining Gx and Gy, test many orientations and keep the winner.	**Compass operators**
Must we use derivatives at all?	Comparing brightness inside a small disc needs no differencing.	**SUSAN**
Is this an edge or a corner?	One number per pixel cannot tell them apart. Two can.	**Structure tensor**
Which scale is the right one?	Fine detail and coarse structure are both real. Look at several.	**Scale space**
Which detector is actually better?	"Looks good" is not a measurement.	**Precision–recall**

The Scharr Operator

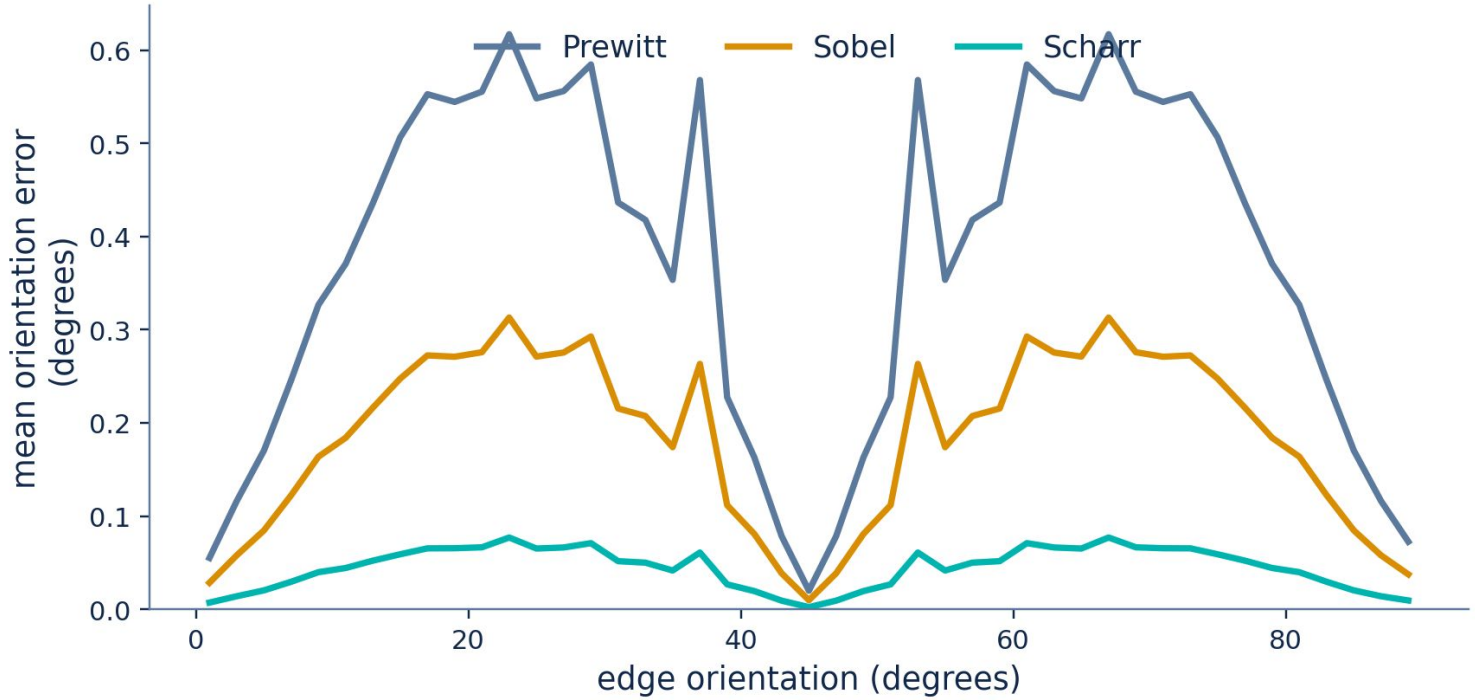
Scharr Gx

-3	0	3
-10	0	10
-3	0	3

Scharr Gy

-3	-10	-3
0	0	0
3	10	3

(coefficients shown × 32)



Sobel's 1-2-1 weighting is a reasonable guess; Scharr's 3-10-3 is the result of optimizing for rotational symmetry. Measured on a radial test pattern, the mean error in estimated edge direction drops from about 0.38° (Prewitt) to 0.19° (Sobel) to 0.05° (Scharr) — for identical cost.

Compass Operators (Kirsch)

0°

-3	-3	5
-3	0	5
-3	-3	5

45°

-3	-3	-3
-3	0	5
-3	5	5

90°

-3	-3	-3
-3	0	-3
5	5	5

135°

-3	-3	-3
5	0	-3
5	5	-3

180°

5	-3	-3
5	0	-3
5	-3	-3

225°

5	5	-3
5	0	-3
-3	-3	-3

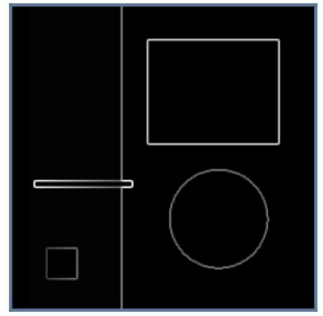
270°

5	5	5
-3	0	-3
-3	-3	-3

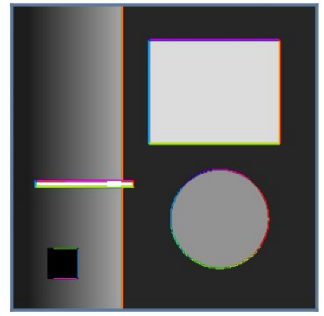
315°

-3	5	5
-3	0	5
-3	-3	-3

max over all 8



direction of the winner



Each mask is the same template rotated by 45°. The strongest response gives both edge strength and edge direction in one step.

Rather than estimating two components and combining them, apply one directional template rotated into eight orientations and keep the largest response. Edge strength and a quantized edge direction fall out together.

NO DERIVATIVES AT ALL

SUSAN: Counting Similar Neighbours

Place a small circular mask on a pixel. Count what fraction of the disc has nearly the same brightness as the centre. That fraction alone tells you where you are.

Flat region → ~100%

Almost the whole disc matches the centre. Nothing interesting here.

Straight edge → ~50%

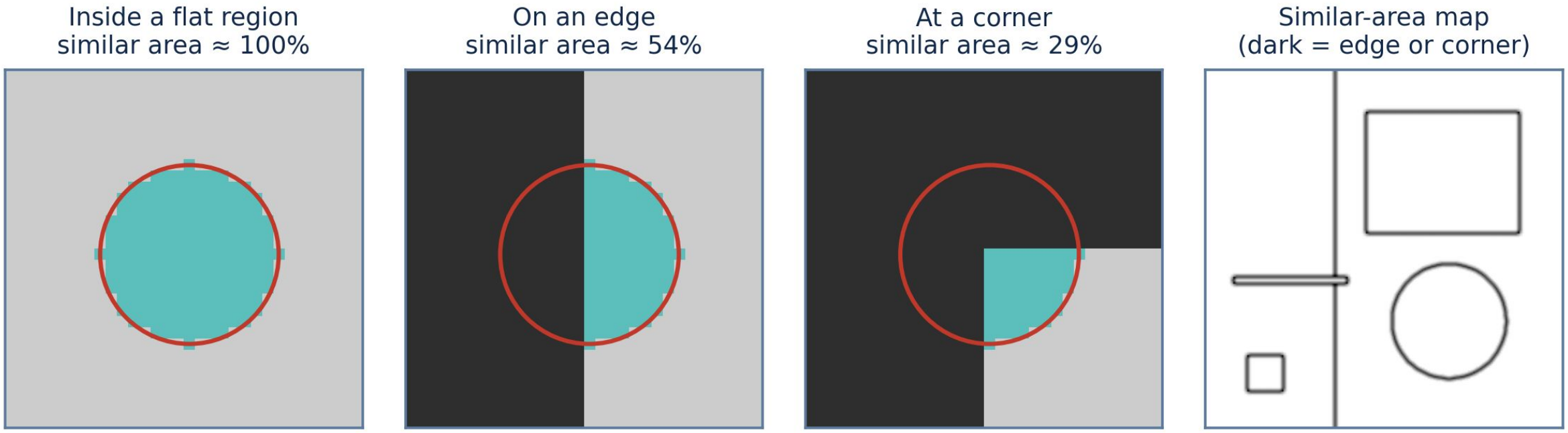
The disc is cut roughly in half by the boundary, so about half of it matches.

Corner → ~25%

Two boundaries cut into the disc, leaving only a wedge that matches. The smaller the matching area, the sharper the feature.

Smith & Brady, 1997 — the same paper that contributed to the bilateral filter in Lecture 2.

SUSAN in Pictures



The teal region is the part of the disc that matches the centre pixel. Its measured area falls from 100% in the flat region to 54% on the edge to 29% at the corner — exactly the behaviour the method is built on. No subtraction of neighbouring pixels is involved, which makes it notably robust to noise.

The Structure Tensor

Instead of one gradient per pixel, collect the gradients over a small window and ask in how many directions the image is changing.

$$J = \text{average of } \begin{bmatrix} G_x^2 & G_xG_y \\ G_xG_y & G_y^2 \end{bmatrix}$$

Both eigenvalues small

Nothing is changing in any direction — a flat region.

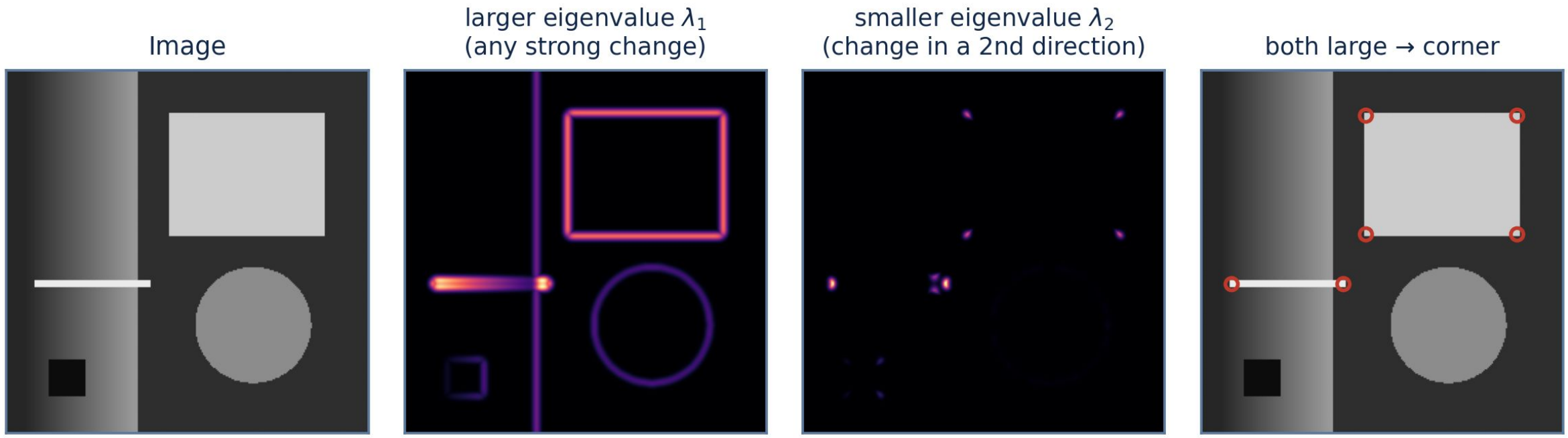
One large, one small

Strong change in a single direction only — an edge.

Both large

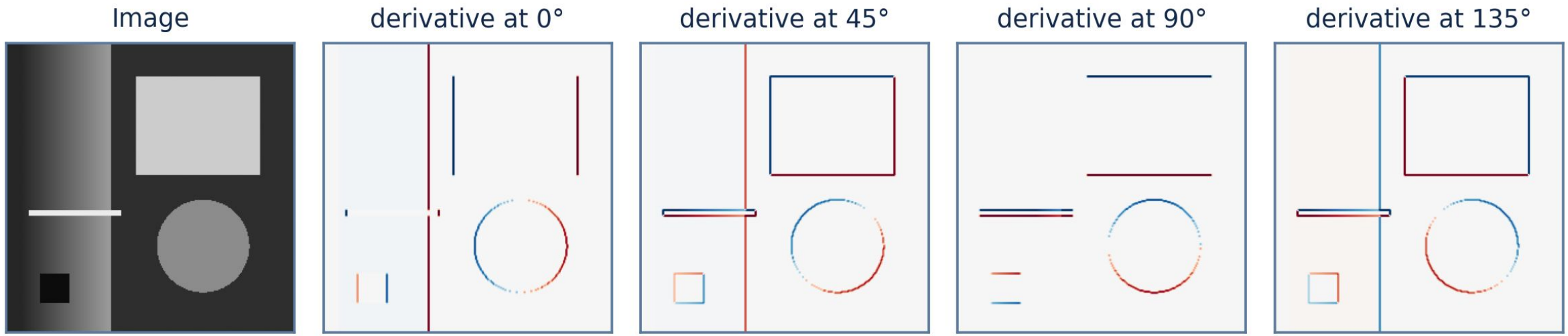
Change in two independent directions at once — a corner. This is the Harris corner detector.

Eigenvalues Separate Edges from Corners



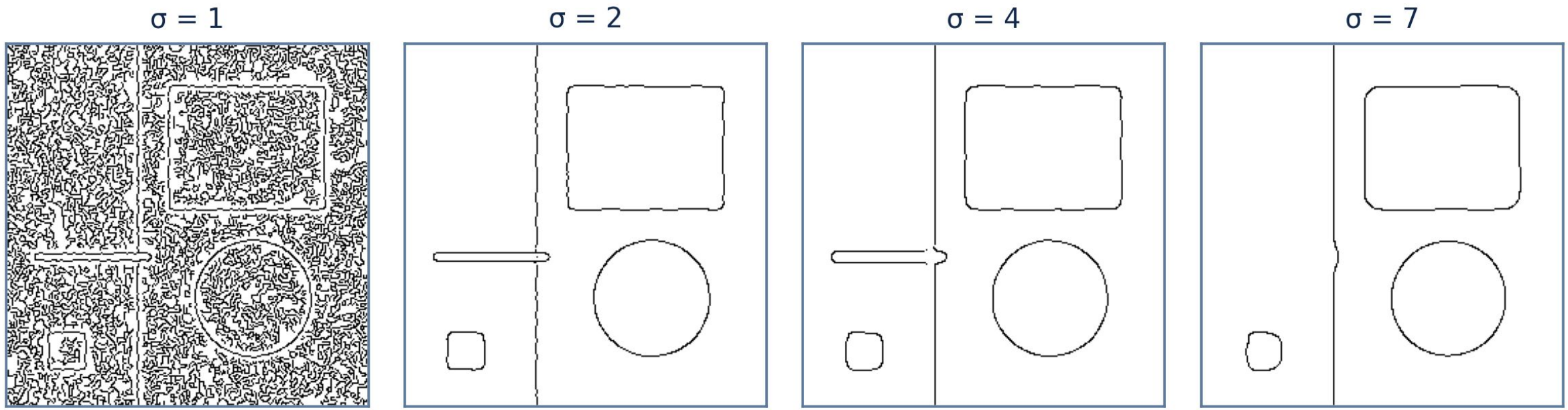
The larger eigenvalue lights up along every boundary. The smaller one is near zero along straight edges and rises only where two boundaries meet — so thresholding it picks out precisely the corners of the square, the ends of the line, and nothing else.

Steerable Filters



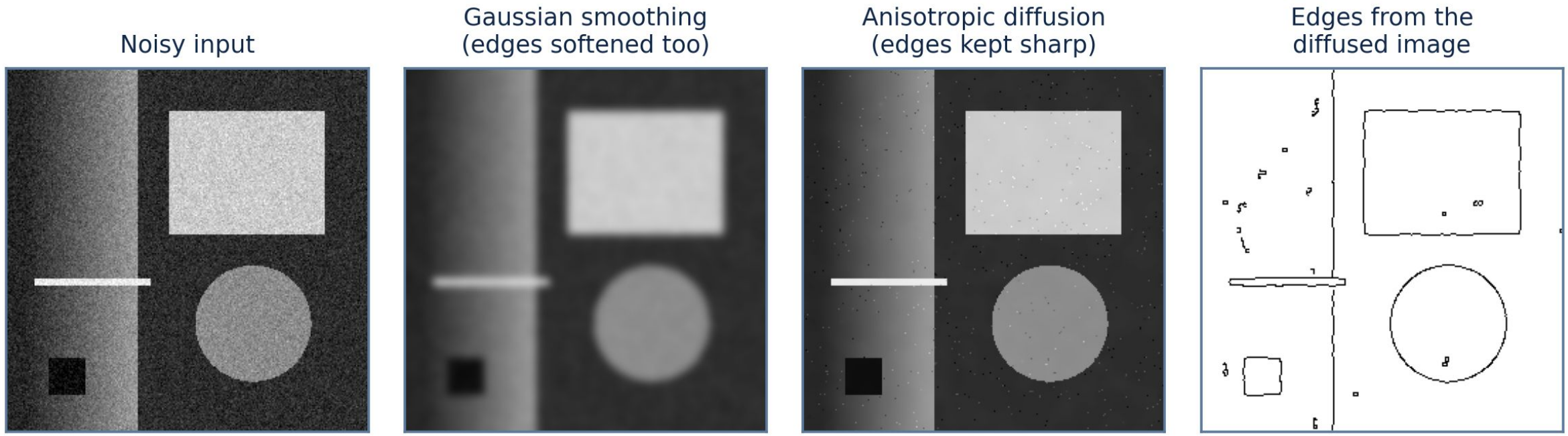
The derivative at any angle θ is just cos θ · Gx + sin θ · Gy. So two filter results are enough to synthesize the response at every orientation — no need for a bank of rotated masks as the compass operators use. The same idea extends to higher-order and more elaborate filters.

Edges Depend on the Scale You Look At



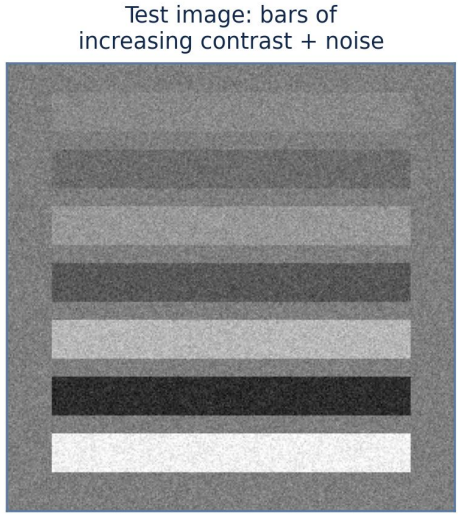
The same image, the same detector, four values of σ. There is no single correct answer: fine texture is real, and so is the coarse outline. Multi-scale methods run several σ and keep the edges that persist across them.

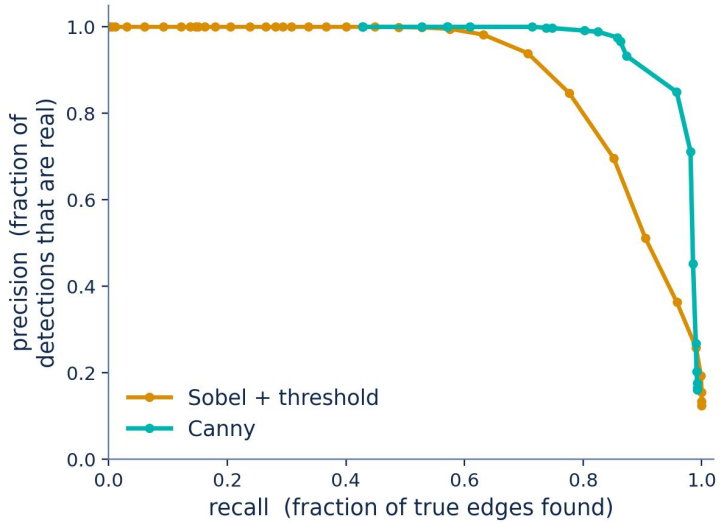
Smoothing That Knows About Edges



Canny's first step blurs edges along with the noise. Perona–Malik diffusion repeatedly averages each pixel with its neighbours, but weights each neighbour down when the difference across it is large — so noise disappears while boundaries stay sharp, and the edges that follow are cleaner.

Measuring an Edge Detector





Precision

Of the pixels called edges, what fraction really are edges? Low precision means clutter.

Recall

Of the real edges, what fraction were found? Low recall means missed boundaries.

A detector is a curve

Raising the threshold trades recall for precision. Canny sits above Sobel throughout — a measurement, not an impression.

Where the Field Went Next

Every detector so far decides using a fixed rule applied to a small window. The modern alternative is to learn the rule from images humans have annotated.

Brightness alone is not enough

A boundary between two textures of equal average brightness produces almost no gradient — yet a human sees it immediately.

Learned detectors

Structured forests, and later convolutional networks, are trained on human-marked boundaries and combine brightness, colour and texture cues automatically.

What carries over

They still begin with oriented derivative-like filters at several scales. Everything in this lecture remains the first layer of what came after.

PART VI SUMMARY

The Advanced Detectors at a Glance

Method	Core idea	Gives you
Scharr	coefficients optimized for rotational symmetry	far more accurate edge direction, same cost
Compass (Kirsch)	one template rotated into 8 orientations, keep the max	strength and direction in one step
SUSAN	count neighbours of similar brightness in a disc	edges and corners without any differencing
Structure tensor / Harris	eigenvalues of the averaged gradient outer product	a way to tell edges from corners
Steerable filters	any orientation from two basis responses	continuous orientation, cheaply
Scale space	run the detector at several σ	edges labelled by the scale they live at
Anisotropic diffusion	smooth, but not across large differences	noise removal that spares the edges

Sharpening and Edge Operators at a Glance

Method	Derivative	Mask	Produces
Laplacian	second	3×3, sums to zero	Edge accents to add back to the image
Composite operator	second	3×3, centre 5 or 9	Sharpened image in a single pass
Unsharp / highboost	none — uses a blur	any smoothing filter	Sharpened image, strength set by k
Roberts	first	2×2	Thin but displaced edge response
Prewitt	first	3×3	Centred edge response
Sobel	first	3×3, centre weighted 2	Centred response, less noise
LoG (Marr–Hildreth)	second, after smoothing	one combined mask	Thin closed contours at zero crossings
Canny	first, after smoothing	multi-stage	Single-pixel edge map

SPATIAL FILTERING, COMPLETE

Where We Have Arrived

Intensity transformations

One pixel at a time.
s = T(r).

Smoothing

Average a neighbourhood.
Keeps slow variation.

Sharpening

Difference a neighbourhood.
Keeps fast variation.

Edge detection

Smooth, differentiate, decide.
A map of boundaries.

Everything in these three lectures is the same operation: slide a window, combine what is inside it. Only the combination rule changed.

THAT COMPLETES SPATIAL FILTERING

Next: the Frequency Domain

The same filtering, described a second way — and some things that are far easier to say there than here.

Questions?